



2011 Trial Examination

FORM VI

MATHEMATICS EXTENSION 1

Wednesday 10th August 2011

General Instructions

- Reading time — 5 minutes
- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.
- All necessary working should be shown in every question.
- Start each question in a new booklet.

Structure of the paper

- Total marks — 84
- All seven questions may be attempted.
- All seven questions are of equal value.

Collection

- Write your candidate number clearly on each booklet.
- Hand in the seven questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place the question paper inside your answer booklet for Question One.

Checklist

- SGS booklets — 7 per boy
- Candidature — 126 boys

Examiner

LYL

QUESTION ONE (12 marks) Use a separate writing booklet.

Marks

- (a) Simplify $\frac{(n+1)!}{n!}$. **1**
- (b) Find $\int \frac{1}{9+x^2} dx$. **1**
- (c) When the polynomial $P(x) = x^3 + 3x^2 + ax - 10$ is divided by $x - 2$, the remainder is 24. Find a . **2**
- (d) Differentiate $y = \sin^{-1}(x^3)$. **2**
- (e) Suppose that α , β and γ are the roots of the equation $x^3 - 3x^2 - 4x + 12 = 0$.
- (i) Write down the value of $\alpha\beta + \alpha\gamma + \beta\gamma$. **1**
- (ii) Hence find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. **1**
- (f) (i) Without the use of calculus, sketch the polynomial $y = x(x+1)(x-4)$ showing all the intercepts with the axes. **2**
- (ii) Hence, or otherwise, solve the inequation $\frac{x(x+1)}{x-4} \geq 0$. **2**

QUESTION TWO (12 marks) Use a separate writing booklet.

Marks

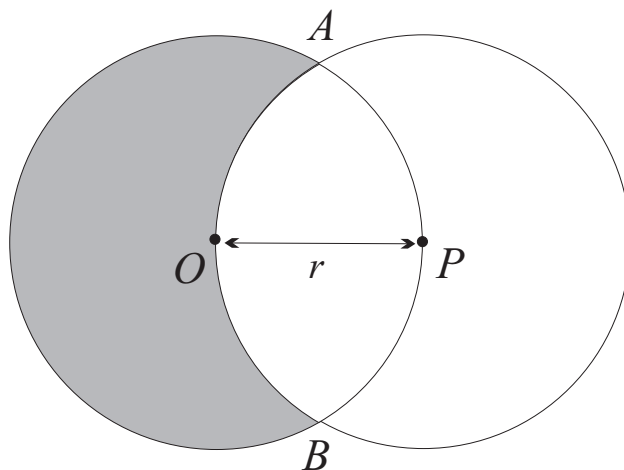
- (a) Find the exact value of $\sin^{-1}(\sin \frac{2\pi}{3})$. **1**
- (b) Find $\lim_{x \rightarrow \infty} \frac{3-x}{2x+3}$. **1**
- (c) The point A is $(2, -4)$ and the point B is $(5, 2)$. The point P divides the interval AB externally in the ratio $4:1$. Find the coordinates of P . **2**
- (d) Find the gradient of the tangent to the curve $y = \tan^{-1}(\sin x)$ at $x = \pi$. **2**
- (e) A ball is projected vertically upwards from the ground. After t seconds, the height of the ball is given by $h = 45t - 5t^2$ metres.
- (i) At what time does the ball returns to the ground? **1**
- (ii) When is the ball instantaneously at rest? **1**
- (iii) What is the greatest height attained by the ball? **1**
- (f) (i) Sketch the graph of the function $y = |x^2 - 4|$. **2**
- (ii) At what points is $f(x) = |x^2 - 4|$ not differentiable? **1**

QUESTION THREE (12 marks) Use a separate writing booklet.**Marks**

- (a) State the domain and range of
- $f(x) = 2 \cos^{-1} \frac{x}{4}$
- .

2

(b)



In the diagram above, two circles of equal radius r units are drawn such that their centres O and P are r units apart. The two circles intersect at A and B .

- (i) Show that the quadrilateral $AOBP$ is a rhombus. **1**
- (ii) Show that $\angle AOB = 120^\circ$. **1**
- (iii) Find the area of the shaded region in terms of r . **2**
- (c) The function $f(x) = x \log x + x - 1.1$ has a zero near $x = 1$. Take $x = 1$ as a first approximation and use Newton's method once to obtain a closer approximation to this zero. **3**
- (d) Find the term independent of x in the expansion of $\left(4x^3 - \frac{1}{x}\right)^{12}$. **3**

QUESTION FOUR (12 marks) Use a separate writing booklet.**Marks**

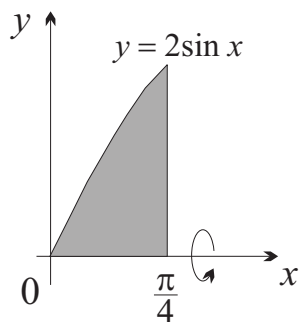
- (a) Given that
- α
- is an acute angle and
- $\cos \alpha = \frac{3}{4}$
- , find the exact value of
- $\tan \frac{\alpha}{2}$
- .

2

- (b) Using the substitution
- $u = 4x + 1$
- , evaluate
- $\int_0^1 \frac{4x}{(4x+1)^2} dx$
- .

3

(c)



The diagram above shows the region bounded by the curve $y = 2\sin x$, the x -axis and the line $x = \frac{\pi}{4}$. Find the exact volume of the solid generated when the shaded region is rotated about the x -axis.

3

- (d) A particle is moving in a straight line according to the equation

$$x = \sqrt{3} \cos 3t - \sin 3t,$$

where x metres is its displacement from the origin after t seconds.

- (i) Show that the particle is moving in simple harmonic motion.

2

- (ii) Find the time at which the particle first passes through the origin.

2

QUESTION FIVE (12 marks) Use a separate writing booklet.

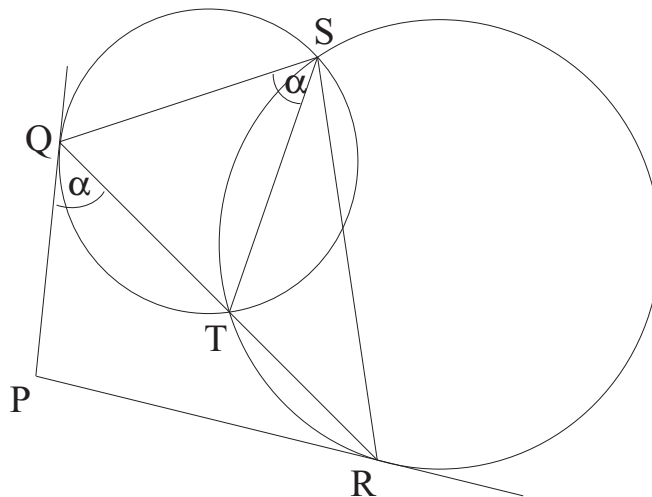
Marks

4

- (a) Prove by mathematical induction that for all positive integer values of n ,

$$\frac{1}{3} \times \frac{1}{1} + \frac{1}{5} \times \frac{1}{3} + \frac{1}{7} \times \frac{1}{5} + \cdots + \frac{1}{(2n+1)} \times \frac{1}{(2n-1)} = \frac{n}{2n+1}.$$

- (b)



In the diagram above PQ and PR are tangents to the circles SQT and STR respectively, and the points Q , T and R are collinear.

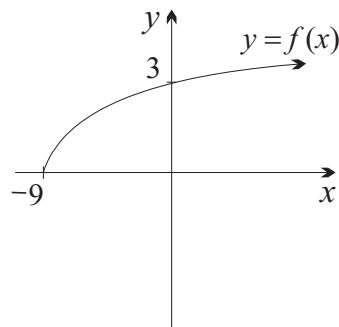
- (i) Given that $\angle QST = \alpha$, state a reason why $\angle PQT = \alpha$.

1

- (ii) Prove that $PQSR$ is a cyclic quadrilateral.

2

- (c)



The diagram above shows a sketch of $y = f(x)$ where $f(x) = \sqrt{x+9}$.

- (i) Copy the diagram. On the same set of axes, sketch the graph of the inverse function $y = f^{-1}(x)$, clearly marking the x and y -intercepts.

1

- (ii) What is the domain of $f^{-1}(x)$?

1

- (iii) Find an expression for $f^{-1}(x)$.

1

- (iv) Given that the graphs of $y = f(x)$ and $y = f^{-1}(x)$ meet at the point P , find the x -coordinate of P .

2

Exam continues next page ...

QUESTION SIX (12 marks) Use a separate writing booklet.**Marks**

- (a) When an object falls from rest at $t = 0$ through a resisting liquid, the rate of change of its velocity at time t is given by $\frac{dv}{dt} = -k(v - 600)$, where k is a positive constant.

(i) Show that $v = 600 + Pe^{-kt}$ is a solution to the differential equation for some constant P . **1**

(ii) If the velocity of the object at $t = 3$ s is 25 ms^{-1} , find P and k . **2**

(iii) Find the velocity of the object at $t = 10$ s. Give your answer correct to one decimal place. **1**

(iv) What is the limiting value of v as $t \rightarrow \infty$? **1**

- (b) Let $(2x + y)^{12} = \sum_{k=0}^{12} T_k$ where $T_k = {}^{12}C_k \times (2x)^{12-k} \times y^k$.

(i) Show that $\frac{T_{k+1}}{T_k} = \frac{y(12-k)}{2x(k+1)}$. **1**

(ii) Suppose that $x = 4$ and $y = 5$ in the expansion of $(2x + y)^{12}$. Show that there are two consecutive terms that are equal, and greater in value than any of the other terms. **2**

- (c) (i) Find the general solutions of the equation **3**

$$2 \cos 3x \sin 4x + 2 \cos 3x - \sin 4x - 1 = 0.$$

(ii) Hence write down all the solutions in the domain $0 \leq x \leq \pi$. **1**

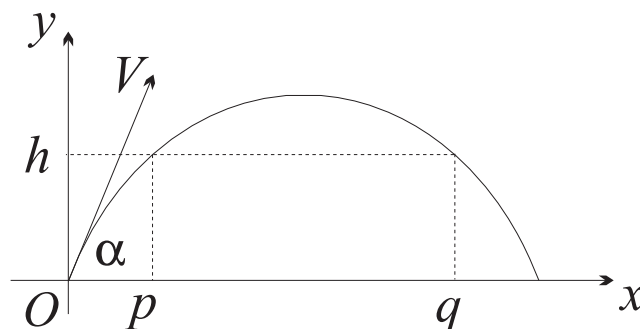
QUESTION SEVEN (12 marks) Use a separate writing booklet.**Marks**

- (a) Using the identity
- $(1+x)^{2n} = (1+x)^n(1+x)^n$
- , show that

2

$$\binom{2n}{n} = \sum_{k=0}^n \binom{n}{k}^2.$$

- (b)



A particle is projected from a point O at an angle of elevation α with level ground at an initial velocity $V \text{ ms}^{-1}$, as in the diagram above.

The particle just clears two vertical poles of height h metres at horizontal distances of p and q metres from O . Take acceleration due to gravity as 10 ms^{-2} and ignore air resistance. You may assume the equations of motion:

$$x = Vt \cos \alpha$$

$$y = Vt \sin \alpha - 5t^2$$

- (i) Find an expression for
- V^2
- in terms of
- α
- ,
- p
- and
- h
- .

2

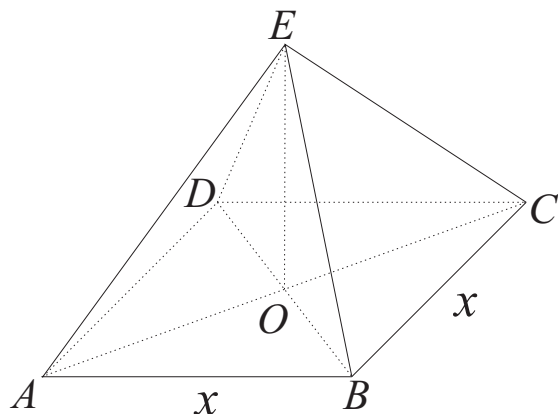
- (ii) Hence show that
- $\tan \alpha = \frac{h(p+q)}{pq}$
- .

2

Question Seven continues on the next page

QUESTION SEVEN (Continued)

(c)



A square pyramid has its apex vertically above the centre of the base. The square base has side length x and the volume of the pyramid is V . The area of each triangular face is $\frac{S}{4}$ for some constant S .

(i) Show that $S^2 = x^4 + \frac{36V^2}{x^2}$. **2**

(ii) Prove that if V is constant and x is variable, then S has its minimum value when **2**

$$x^3 = (3\sqrt{2})V.$$

(iii) When S is at its minimum, show that each triangular face is equilateral. **2**

END OF EXAMINATION

B L A N K P A G E

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

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Question 1

a) $\frac{(n+1)!}{n!} = \underline{n+1}$

b) $\int \frac{dx}{9+x^2} = \underline{\underline{\frac{1}{3} \tan^{-1} \frac{x}{3} + c}}$

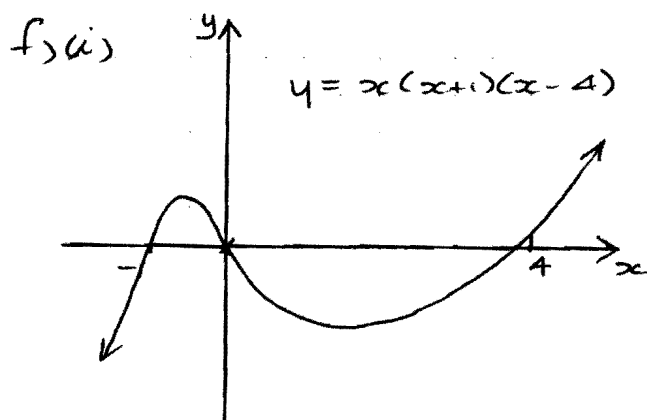
c) $P(2) = 24$
 $2^3 + 3(2)^2 + a(2) - 10 = 24$
 $8 + 12 + 2a - 10 = 24$
 $2a = 14$
 $\underline{a = 7}$

d) $y = \sin^{-1} x^3$
 $\frac{dy}{dx} = \frac{3x^2}{\sqrt{1-x^6}}$

e) $x^3 - 3x^2 - 4x + 12 = 0$

(i) $\alpha\beta + \alpha\gamma + \beta\gamma = \underline{-4}$

(ii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma}$
 $= \frac{-4}{-12}$
 $= \underline{\underline{\frac{1}{3}}}$



(ii) $\frac{x(x+1)}{x-4} \geq 0$
 $\frac{x(x+1)(x-4)^2}{(x-4)} \geq 0, x \neq 4$
 $x(x+1)(x-4) \geq 0$
 $\underline{\underline{-1 \leq x \leq 0, x > 4}}$

Question 2

a) $\sin^{-1}(\sin \frac{2\pi}{3}) = \sin^{-1} \frac{\sqrt{3}}{2}$
 $= \underline{\underline{\frac{\pi}{3}}}$

b) $\lim_{x \rightarrow \infty} \frac{3-x}{2x+3} = \underline{\underline{-\frac{1}{2}}}$

c) $A(2, -4) \quad B(5, 2)$
 $-4:1$
 $P = \left(\frac{2-20}{-3}, \frac{-4-8}{-3} \right)$
 $= \underline{\underline{(6, 4)}}$

d) $y = \tan^{-1}(\sin x)$

$\frac{dy}{dx} = \frac{\cos x}{1 + \sin^2 x}$
 at $x = \pi$, $\frac{dy}{dx} = \frac{-1}{1+0}$
 $= -1$

$\therefore$ slope of tangent at $x = \pi$ is -1

e) $h = 45t - 5t^2$

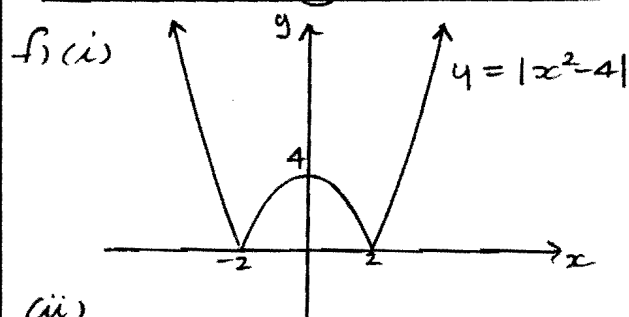
(i) $45t - 5t^2 = 0$
 $5t(9-t) = 0$
 $t = 0$ or $t = 9$

$\therefore$ returns to ground after 9 seconds

(ii) ball will be at rest at greatest height which by symmetry is after $4\frac{1}{2}$ seconds

(iii) $t = \frac{9}{2}$, $h = 45\left(\frac{9}{2}\right) - 5\left(\frac{9}{2}\right)^2$
 $= \underline{\underline{\frac{405}{4}}}$

$\therefore$ greatest height is $101\frac{1}{4}$ m



(ii) function is not differentiable at $x = \pm 2$

Question 3

a) domain: $-1 \leq \frac{x}{2} \leq 1$
 $-4 \leq x \leq 4$

range: $0 \leq \frac{y}{2} \leq \pi$
 $0 \leq y \leq 2\pi$

b) $OA = OB = PB = PA = r$ (= radii)

$\therefore$ $AOBP$ is a rhombus
 (4 = sides)

(ii) $OP = r$ (given)
 $\therefore \triangle AOP$ is equilateral
 (3 = sides)

$\angle AOP = 60^\circ$ ($\angle$ in equilateral $\triangle$)
 Similarly $\angle BOP = 60^\circ$
 $\angle AOB = \angle AOP + \angle BOP$
 $= 120^\circ$

(iii) $A = \pi r^2 - 2 \left[\frac{1}{2} r^2 \theta - \frac{1}{2} r^2 \sin \theta \right]$
 $= r^2 \left[\pi - \theta + \sin \theta \right]$
 $= r^2 \left[\pi - \frac{2\pi}{3} + \frac{\sqrt{3}}{2} \right]$
 $= \frac{r^2 (\pi + 3\sqrt{3})}{6}$

c) $f(x) = x \log x + x - 1.1$

$f'(x) = (x) \left(\frac{1}{x} \right) + (\log x)(1) + 1 - 1.1$
 $= \log x + 0.9$

$x_1 = 1 - \frac{(1) \log(1) + 1 - 1.1}{\log(1) + 0.9}$
 $= 1.9$

d) $T_{k+1} = {}^{12}C_k (4x^3)^{12-k} \left(-\frac{1}{x}\right)^k$
 $x^{36-3k} \cdot x^{-k} = x^0$
 $36 - 4k = 0$
 $k = 9$

$T_{10} = {}^{12}C_9 4^3 (-1)^9$
 $= -14080$

Question 4

a) $\frac{1-x^2}{1+x^2} = \frac{3}{4}$
 $4 - 4x^2 = 3 + 3x^2$
 $7x^2 = 1$
 $x^2 = \frac{1}{7}$
 $x = \pm \frac{1}{\sqrt{7}}$

as α is acute, so is $\frac{\alpha}{2}$

$\therefore \tan \frac{\alpha}{2} = \frac{1}{\sqrt{7}}$

b) $\int_0^1 \frac{4x}{(4x+1)^2} dx$ $u = 4x+1$
 $du = 4dx$
 $x=0, u=1$
 $x=1, u=5$
 $= \frac{1}{4} \int_1^5 \frac{u-1}{u^2} du$
 $= \frac{1}{4} \int_1^5 \left[\frac{1}{u} - \frac{1}{u^2} \right] du$
 $= \frac{1}{4} \left[\log u + \frac{1}{u} \right]_1^5$
 $= \frac{1}{4} \left(\log 5 + \frac{1}{5} - \log 1 - 1 \right)$
 $= \frac{1}{4} \left(\log 5 - \frac{4}{5} \right)$

c) $V = \pi \int_0^{\frac{\pi}{4}} 4 \sin^2 x dx$
 $= 2\pi \int_0^{\frac{\pi}{4}} (1 - \cos 2x) dx$
 $= 2\pi \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{4}}$
 $= 2\pi \left(\frac{\pi}{4} - \frac{1}{2} - 0 + 0 \right)$
 $= \frac{\pi^2 - \pi}{2} \text{ units}^3$

d) (i) $x = \sqrt{3} \cos 3t - \sin 3t$
 $\dot{x} = -3\sqrt{3} \sin 3t - 3 \cos 3t$
 $\ddot{x} = -9\sqrt{3} \cos 3t + 9 \sin 3t$
 $= -9x$
 $\therefore$ particle moves in SHM

(ii) $\sqrt{3} \cos 3t - \sin 3t = 0$

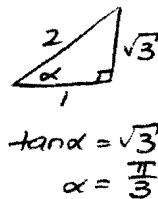
$2 \sin \left(3t - \frac{\pi}{3} \right) = 0$

$3t - \frac{\pi}{3} = 0$

$3t = \frac{\pi}{3}$

$t = \frac{\pi}{9}$

$\therefore$ particle first passes through origin after $\frac{\pi}{9}$ seconds



Question 5

a) $n=1$

$$\text{LHS} = \frac{1}{3} \times \frac{1}{1} = \frac{1}{3}$$

$$\text{RHS} = \frac{1}{2(1)+1} = \frac{1}{3}$$

$$\text{LHS} = \text{RHS}$$

Hence the result is true for $n=1$

Assume the result is true for $n=k$ where k is a positive integer

$$\text{i.e. } \frac{1}{3} \times \frac{1}{1} + \frac{1}{5} \times \frac{1}{3} + \dots + \frac{1}{(2k+1)} \times \frac{1}{(2k-1)} = \frac{k}{2k+1}$$

Prove true for $n=k+1$

i.e. Prove

$$\frac{1}{3} \times \frac{1}{1} + \frac{1}{5} \times \frac{1}{3} + \dots + \frac{1}{(2k+3)} \times \frac{1}{(2k+1)} = \frac{k+1}{2k+3}$$

PROOF

$$\begin{aligned} & \frac{1}{3} \times \frac{1}{1} + \frac{1}{5} \times \frac{1}{3} + \dots + \frac{1}{(2k+1)} \times \frac{1}{(2k-1)} + \frac{1}{(2k+3)} \times \frac{1}{(2k+1)} \\ &= \frac{k}{2k+1} + \frac{1}{(2k+3)} \times \frac{1}{(2k+1)} \\ &= \frac{k(2k+3) + 1}{(2k+3)(2k+1)} \\ &= \frac{2k^2 + 3k + 1}{(2k+3)(2k+1)} \\ &= \frac{(2k+1)(k+1)}{(2k+3)(2k+1)} \\ &= \frac{k+1}{2k+3} \end{aligned}$$

Hence the result is true for $n=k+1$ if it is true for $n=k$.

Since the result is true for $n=1$ then it is true for all positive integral values of n by induction.

b) (i) $\angle PQT = \alpha$ by the alternate segment theorem.

(ii) Let $\angle RST = \beta$
 $\angle TRP = \beta$ (alternate segment theorem)

$$\angle QPR = \angle PQT + \angle TRP = 100^\circ \text{ (Theorem)}$$

$$\therefore \angle QPR = 100 - \alpha - \beta$$

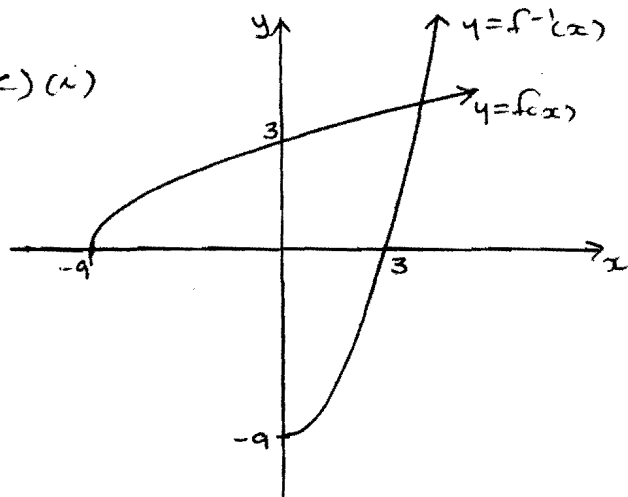
$$\angle QSR = \angle QST + \angle RST \text{ (Common } \angle)$$

$$\angle QSR = \alpha + \beta$$

$$\therefore \angle QPR + \angle QSR = 100^\circ$$

PQSR is a cyclic quadrilateral as opposite $\angle$ s supplementary

c) (i)



(ii) domain $f^{-1}(x)$: $x \geq 0$

$$\begin{aligned} \text{(iii) } f^{-1}: x &= \sqrt{y+9} \\ x^2 &= y+9 \\ y &= x^2 - 9, x \geq 0 \end{aligned}$$

(iv) meet on line $y=x$

$$\begin{aligned} x &= x^2 - 9 \\ x^2 - x - 9 &= 0 \\ x &= \frac{1 \pm \sqrt{37}}{2} \end{aligned}$$

but $x \geq 0$

$\therefore$ x coordinate of P is

$$x = \frac{1 + \sqrt{37}}{2}$$

Question 6

$$\begin{aligned} \text{a) (i) } v &= 600 + Pe^{-kt} \\ \frac{dv}{dt} &= -kPe^{-kt} \\ &= -k(Pe^{-kt} + 600 - 600) \\ &= -k(v - 600) \end{aligned}$$

$$\begin{aligned} \text{(ii) when } t=0, v=0 \\ \text{i.e. } 0 &= 600 + Pe^0 \\ P &= -600 \end{aligned}$$

$$\begin{aligned} \text{when } t=3, v=25 \\ \text{i.e. } 25 &= 600 - 600e^{-3k} \\ 600e^{-3k} &= 575 \\ e^{-3k} &= \frac{575}{600} = \frac{23}{24} \\ -3k &= \log \frac{23}{24} \\ k &= -\frac{1}{3} \log \frac{23}{24} \end{aligned}$$

$$\begin{aligned} \text{(iii) when } t=10, \\ v &= 600 - 600e^{-10k} \\ &= 600 - 600e^{-\frac{10}{3} \log \frac{23}{24}} \\ &= 600 - 600 \left(\frac{23}{24} \right)^{\frac{10}{3}} \\ &= 79.35716261 \dots \\ &= \underline{79.4 \text{ m/s}} \end{aligned}$$

(iv) as $t \rightarrow \infty$, $e^{-kt} \rightarrow 0$

$$\therefore v \rightarrow 600 \text{ m/s}$$

$$\begin{aligned} \text{b) (i)} \quad \frac{T_{k+1}}{T_k} &= \frac{{}^{12}C_{k+1} (2x)^{12-k} y^{k+1}}{{}^{12}C_k (2x)^{12-k} y^k} \\ &= \frac{12!}{(11-k)!(k+1)!} \times \frac{(12-k)!k!}{12!} \times \frac{y}{2x} \\ &= \frac{12-k}{k+1} \times \frac{y}{2x} \\ &= \frac{y(12-k)}{2x(k+1)} \end{aligned}$$

(ii) If $T_{k+1} > T_k$ then T_{k+1} is the greatest term

$$\therefore \frac{T_{k+1}}{T_k} > 1$$

$$\frac{y(12-k)}{2x(k+1)} > 1$$

$$\frac{5(12-k)}{8(k+1)} > 1$$

$$60 - 5k > 8k + 8$$

$$13k \leq 52$$

$$k \leq 4$$

$$\therefore T_4 = {}^{12}C_4 8^8 5^4 \text{ and}$$

$$T_5 = {}^{12}C_5 8^7 5^5 \text{ are both the greatest terms.}$$

$$\text{c) } 2\cos 3x \sin 4x + 2\cos 3x - \sin 4x = 0$$

$$2\cos 3x (\sin 4x + 1) - (\sin 4x + 1) = 0$$

$$(\sin 4x + 1)(2\cos 3x - 1) = 0$$

$$\sin 4x = -1 \text{ or } \cos 3x = \frac{1}{2}$$

$$4x = \pi k + (-1)^k \sin^{-1}(-1)$$

$$\text{or } 3x = 2\pi k \pm \cos^{-1} \frac{1}{2}$$

where k is an integer

$$4x = \pi k + (-1)^k \frac{3\pi}{2} \text{ or } 3x = 2\pi k \pm \frac{\pi}{3}$$

$$= 2\pi k + \frac{3\pi}{2}$$

$$= \frac{\pi(3+4k)}{2}$$

$$= \frac{\pi(6k \pm 1)}{3}$$

$$x = \frac{\pi(3+4k)}{8} \text{ or } x = \frac{\pi(6k \pm 1)}{9}$$

where k is an integer

$$\text{(ii)} \quad \frac{3\pi}{8}, \frac{7\pi}{8}, \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$$

Question 7

a) coefficient of x^n in $(1+2x)^{2n}$

$$= \binom{2n}{n}$$

coefficient of x^n in $(1+x)^n (1+x)^n$

$$\left[\binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n-1}x^{n-1} + \binom{n}{n}x^n \right]^2$$

coefficient of x^n

$$= \binom{n}{0}\binom{n}{n} + \binom{n}{1}\binom{n}{n-1} + \dots + \binom{n}{n-1}\binom{n}{1} + \binom{n}{n}\binom{n}{0}$$

however

$$\binom{n}{k} = \binom{n}{n-k}$$

$$= \binom{n}{0}^2 + \binom{n}{1}^2 + \dots + \binom{n}{n-1}^2 + \binom{n}{n}^2$$

$$\therefore \binom{2n}{n} = \sum_{k=0}^n \binom{n}{k}^2$$

b) $x = vt \cos \alpha$

$$t = \frac{x}{v \cos \alpha}$$

$$y = Vt \sin \alpha - 5t^2$$

$$= x \tan \alpha - \frac{5x^2}{v^2 \cos^2 \alpha}$$

$$\therefore h = p \tan \alpha - \frac{5p^2}{v^2 \cos^2 \alpha}$$

$$\frac{5p^2}{v^2 \cos^2 \alpha} = p \tan \alpha - h$$

$$v^2 \cos^2 \alpha = \frac{5p^2}{p \tan \alpha - h}$$

$$v^2 = \frac{5p^2}{\cos^2 \alpha (p \tan \alpha - h)}$$

(ii) similarly

$$v^2 = \frac{5q^2}{\cos^2 \alpha (q \tan \alpha - h)}$$

$$\therefore \frac{5p^2}{\cos^2 \alpha (p \tan \alpha - h)} = \frac{5q^2}{\cos^2 \alpha (q \tan \alpha - h)}$$

$$p^2 (q \tan \alpha - h) = q^2 (p \tan \alpha - h)$$

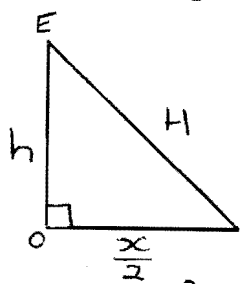
$$(p^2 q - pq^2) \tan \alpha = hp^2 - hq^2$$

$$\tan \alpha = \frac{h(p+q)(p-q)}{pq(p-q)}$$

$$= \frac{h(p+q)}{pq}$$

$$c) V = \frac{1}{3} x^2 h$$

$$h = \frac{3V}{x^2}$$



$$H^2 = \frac{9V^2}{x^4} + \frac{x^2}{4}$$

$$\frac{S}{4} = \frac{1}{2} x H$$

$$S = 2xH$$

$$S^2 = 4x^2 H^2$$

$$= 4x^2 \left(\frac{9V^2}{x^4} + \frac{x^2}{4} \right)$$

$$= \underline{\underline{\frac{36V^2}{x^2} + x^4}}$$

$$(ii) \frac{dS^2}{dx} = -\frac{72V^2}{x^3} + 4x^3$$

stationary pts occur when $\frac{dS^2}{dx} = 0$

$$\text{i.e. } 4x^3 = \frac{72V^2}{x^3}$$

$$(x^3)^2 = 18V^2$$

$$x^3 = 3\sqrt{2} V$$

x^3	4 V	$3\sqrt{2} V$	$5V$
$\frac{dS^2}{dx}$	$-2V$	0	$\frac{28}{3}V$

∴ when $x^3 = 3\sqrt{2} V$, S has a minimum value.

$$(iii) V = \frac{x^3}{3\sqrt{2}}$$

$$\therefore h = \frac{3x^3}{3\sqrt{2}} \times \frac{1}{x^2}$$

$$= \frac{x}{\sqrt{2}}$$

$$H^2 = \frac{x^2}{2} + \frac{x^2}{4}$$

$$= \frac{3x^2}{4}$$

$$H = \frac{\sqrt{3}x}{2}$$

$$\tan \angle ECB = \frac{H}{\frac{x}{2}}$$

$$= \frac{\sqrt{3}x}{2} \times \frac{2}{x}$$

$$= \sqrt{3}$$

$$\angle ECB = 60^\circ$$

$$\angle ECB = \angle EBC$$

$$\therefore \angle EBC = 60^\circ$$

Thus $\triangle EBC$ is equilateral.